CampusNode LMS – Installation Guide

Project: CampusNode Learning Management System
Version: 1.0
Intended audience: School IT technician / system administrator
Deployment type: Local console application

1. Purpose

This guide explains how to prepare a school computer, build and launch CampusNode LMS, and complete its initial setup.

CampusNode LMS is a C++17 console application that uses SQLite for persistent data storage and libsodium for password hashing. It can operate locally without requiring an internet connection during normal use.

2. System requirements

The installation computer should have:

- Windows or a compatible Linux operating system.
- A C++17-compatible compiler.
- CMake build system.
- SQLite 3 development/runtime files.
- libsodium development/runtime files.
- Sufficient disk space for the application, database and backups.
- Permission to create and modify files in the installation directory.

The application must be built with its required dependencies available to CMake and the compiler.

3. Installation procedure

Step 1: Obtain the application source code

Download or clone the CampusNode LMS repository:

```
https://github.com/King-Eazii/CampusNode-LMS
```

If Git is installed, open PowerShell or a terminal and run:

```
git clone https://github.com/King-Eazii/CampusNode-LMS.git
cd CampusNode-LMS
```

Alternatively, download the repository as a ZIP file, extract it and open a terminal in the extracted project directory.

Step 2: Install the dependencies

Ensure that the computer has the following installed and configured:

1. A C++ compiler supporting C++17.
2. CMake.
3. SQLite 3.
4. libsodium.

The compiler and CMake must be accessible to the terminal. The SQLite and libsodium development files must also be discoverable by the build configuration.

Step 3: Configure the project

From the root directory of the project, run:

```
cmake -S . -B build -DBUILD_TESTING=ON
```

This instructs CMake to configure the project and generate build files in the `build` directory.

Step 4: Build the application

Run:

```
cmake --build build --config Debug
```

The build process compiles the source code and produces the application executable.

For the Windows build configuration used during development, the executable is located at:

```
build\Debug\lms.exe
```

Step 5: Launch the application

From the project root, run:

```
.\build\Debug\lms.exe
```

The application should display the login prompt:

```
=== CampusNode LMS: log in ===
Username (empty to quit):
```

If the application does not start, check that the required runtime libraries are available and that the build completed successfully.

4. First-time setup

On its first startup:

1. CampusNode LMS automatically creates its SQLite database and the required tables.
2. The first administrator account is created

Deployment Diagram

